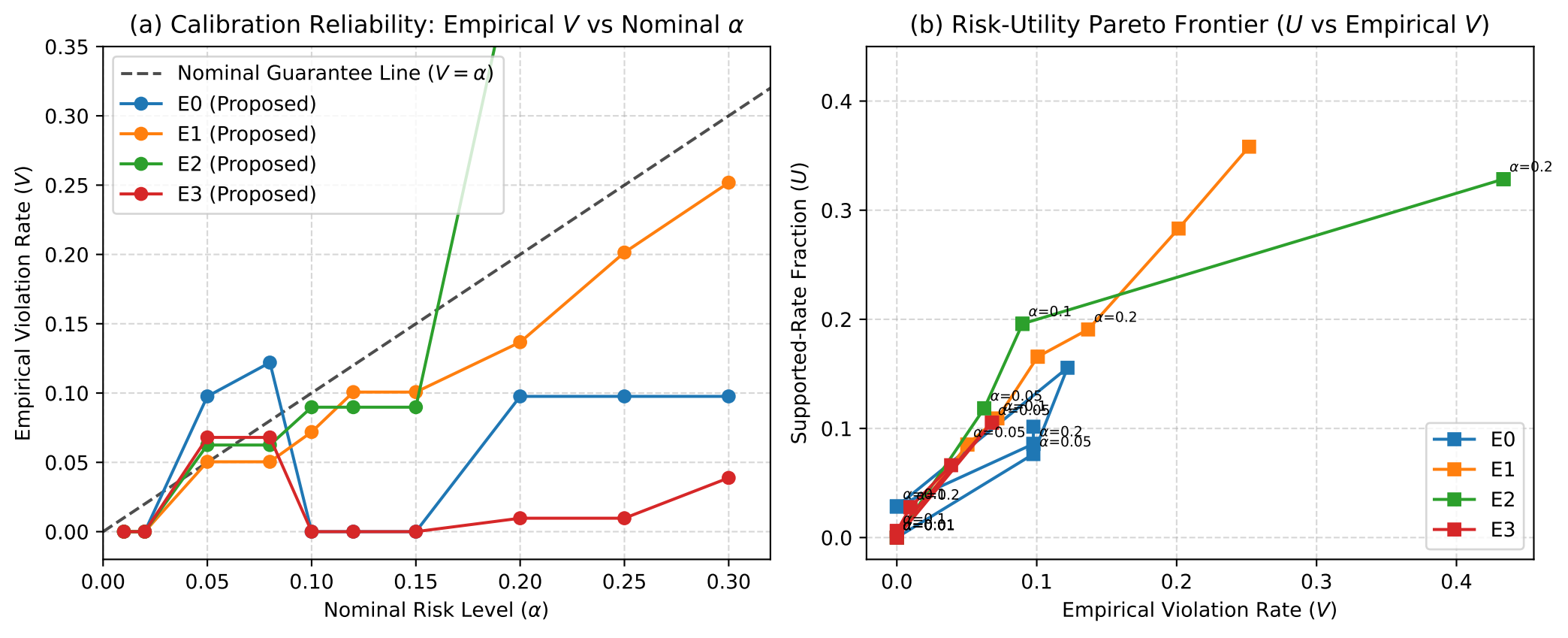
(a) Calibration Reliability: Empirical $V$ vs Nominal $\alpha$
Nominal Guarantee Line ($V = \alpha$)
E0 (Proposed)
E1 (Proposed)
E2 (Proposed)
E3 (Proposed)
Empirical Violation Rate ($V$)
Nominal Risk Level ($\alpha$)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
(b) Risk-Utility Pareto Frontier ($U$ vs Empirical $V$)
Supported-Rate Fraction ($U$)
Empirical Violation Rate ($V$)
0.0
0.1
0.2
0.3
0.4
E0
E1
E2
E3
α=0.2
α=0.1
α=0.2
α=0.05
α=0.05
α=0.05
α=0.2
α=0.05
α=0.2
α=0.1
α=0.01